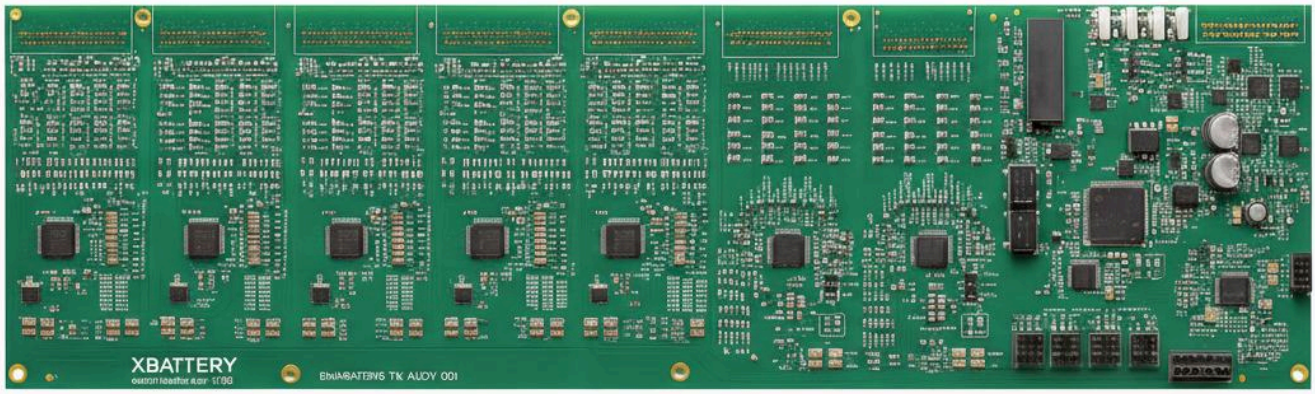
XBATTERY

BharatBMS for EVs



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XBATTERY

Xbattery BharatBMS EV

Xbattery BharatBMS EV is a high-performance BMS designed for electric cars, trucks, and buses. Supporting up to 250S LFP and NMC cells, it ensures accurate monitoring, intelligent balancing, and advanced protection. With CAN FD, RS-485, and Ethernet communication, and compliance with ISO 26262, ASIL-C, AIS-058 and AIS-004(part 3) standards, it delivers automotive-grade safety, reliability, and performance for modern electric mobility.

Applications

EV Car

Efficient power for smart mobility.

EV Truck

Reliable energy for heavy transport.

EV Bus

Safe, scalable fleet performance.

EV 3 Wheeler

Optimized range for urban drive.

XBATTERY

Core Features

Cell chemistry

Compatible with both LFP and NMC chemistries

High-Voltage Ready

Supports up to 186S and up to 600V.

Strong Current Handling

Delivers >1C continuous and >2C peak discharge for high-performance EV powertrains.

Accurate Monitoring

±2mV voltage precision ensures consistent cell health and optimized pack performance.

Smart Balancing

Passive and configurable active balancing up to 300mA keeps every cell perfectly aligned.

Thermal Protection

Tracks up to 32 temperature sensors for safe operation across all climates.

Comprehensive Safety

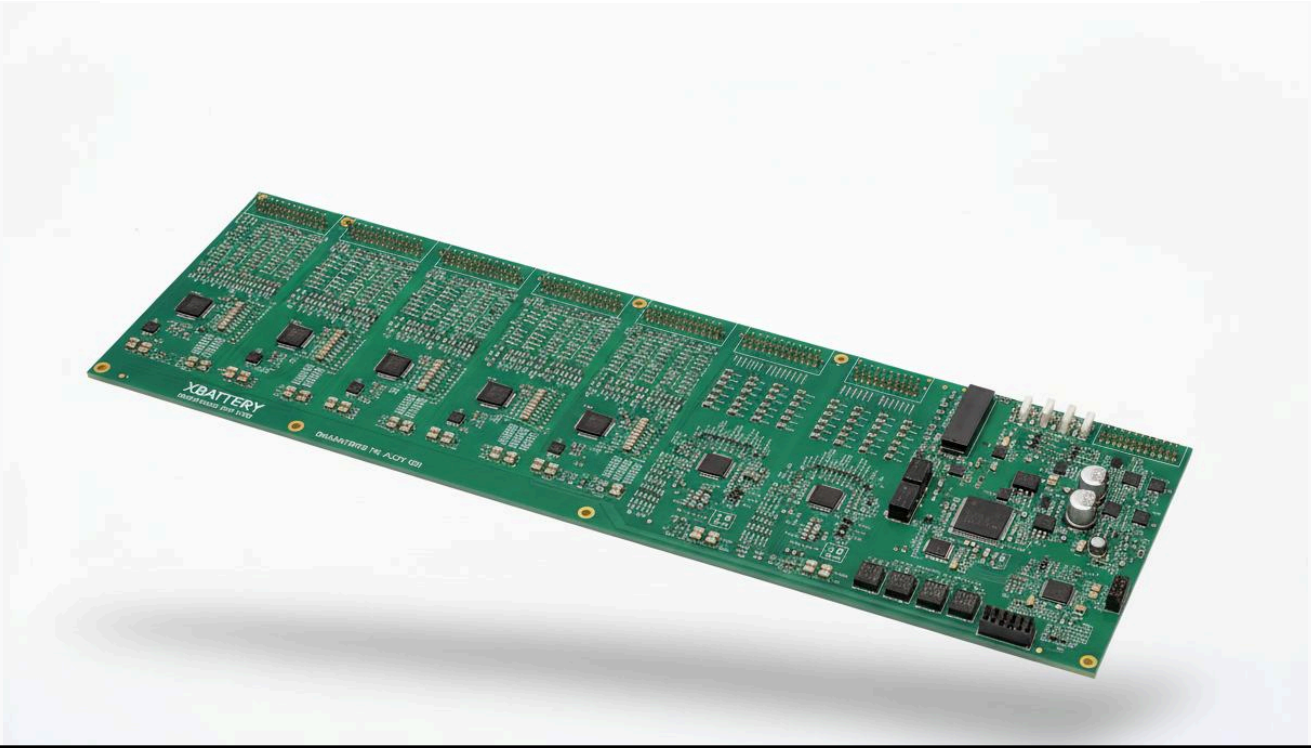
Built-in protection against over-voltage, under-voltage, over-current, short-circuit, and ground faults.

Smart Software Suite

PC GUI and Android App for live monitoring, easy configuration, and firmware upgrades

Advanced Communication

CAN 2.0B, CAN FD, RS-485, ETPL, UART, and optional Ethernet for seamless integration and real-time diagnostics.



XBATTERY

EV BMS Offerings

01

Xbattery BharatBMS-EV-110V

Optimized for e-rickshaws, autos, and tempos. Supports up to 36S configurations, ensuring efficient, safe, and reliable performance for 3-wheel electric vehicles.

02

Xbattery BharatBMS-EV-400V

Designed for passenger cars, hatchbacks, and sedans. Supports up to 125S, delivering advanced power management and enhanced safety for smooth EV operation.

03

Xbattery BharatBMS-EV-600V

Built for SUVs and light commercial vehicles. Supports up to 186S, providing high-performance control, protection, and durability for demanding applications.

Product Specifications

Specification	EV 110V	EV 400V	EV 600V
Cell Configuration	Up to 36S	Up to 125S	Up to 186S
Nominal Pack Voltage Range	110.0V	400.0V	600.0V
Max Pack Voltage Range	216.0V	450.0V	561.6V
Min Pack Voltage Range	150.0V	312.5V	390.0V
Battery Capacity Range	25kWh	80kWh	120kWh
Vehicle Applications	E-Rickshaw, Auto, Tempo	Passenger Cars, Semi Truck	Bus, Truck, Heavy Vehicles

XBATTERY

Tech Specifications

Current Handling	Parameters
Continuous Charge Current Limit	1C(165Ah, 280Ah, ~300Ah)
Peak Charge & Discharge Currents (duration and recovery conditions)	Discharge Currents >1C Continuous , >2C (330Ah, 560Ah, ~600Ah) for up to 30s
Protections & Safety	Auto-recover when within safe limits

Cell Voltage & Balancing	
Cell Voltage Measurement Range	0-5(V)
Measurement Accuracy & Resolution	±2mV Accuracy, 1mV Resolution
Balancing Topology (active or passive/dissipative)	Supported: Passive (Resistor bleed)
Balancing Current	~ Up to 300mA
Balance Enable/Threshold Voltages	Enable >3.40V, ΔV >15mV; stop <5mV

Temperature Sensing & Thermal Management	
Number & Type of Temperature Sensors Supported	Up to 16 × NTC (10kΩ typical)
Sensor Range & Accuracy	−40°C to +125°C, ±1°C typical
Thermal Protection Thresholds and cut-offs	Charge: 0–55°C; Discharge: −20°C to 60°C; Emergency shutdown >80°C

Software, Diagnostics & Tools	
Configuration Software Tools	PC GUI (Windows/Linux), Android app
Event / Error Logging & RTC capabilities	Onboard RTC, >30 days log retention
Versioning & Firmware upgrade method	Over-CAN(ISO-TP), UART fallback

Performance & Lifecycle	
Efficiency	>~98%
Power Consumption (Active / Idle / Sleep current)	Active<5W, Idle<500mW, Sleep <200μA

Protections & Safety	Parameters
Over-voltage Protection (per cell and pack)	Yes
Under-voltage Protection (per cell and pack)	Yes
Over & Under-temperature cut-offs	Yes
Over-current & Short-circuit protection	Yes
Leakage/Ground fault detection	Yes
Protection switching topology	Yes
Precharge circuit or inrush current limiting	Yes

Communication Interfaces	
Supported Protocols	CAN 2.0B, CAN FD, ETPL, RS-485 (Modbus optional), UART (debug), optional Ethernet
Baud Rates & CAN Speeds	125 kbps, 250 kbps, 500 kbps, 1 Mbps; CAN FD up to 2 Mbps
General-purpose I/O	4×Digital Inputs, 4×Digital Outputs
Host Interfaces	CAN-to-PC GUI, Android app, RS-485 adapter, UART_PC_logging

Mechanical & Environmental	
Dimensions & Weight	BMS without case(<150gm),
Mounting Form Factor / Connector Types	Molex, Anderson Powerpole/SB series
Operating & Storage Temperature Range	−40°C to +85°C
Humidity/Environmental Classifications (Indoor/Outdoor, Shock & Vibration specs)	5–95% RH Non-Condensing, Vibration/ Shock
Ingress/IP rating (If enclosure provided)	P20–IP65 depending on enclosure

Standards & Certifications	
Safety Standards	ISO 26262 ASIL-C
EMS/EMC compliance, regulatory notes	AIS-058, AIS-004(part 3)